

2018

COMPUTER SCIENCE – GENERAL

Second Paper

2. (a) Define the term metric in a graph. Prove that the distance between vertices of a connected graph is a metric.
(b) Prove that every circuit has an even number of edges in common with any cut-set
(c) Describe an algorithm with a non-trivial example, for finding a minimum spanning tree from a given weighted undirected graph G. (2+4)+4+6
3. (a) Prove that a connected graph G is an Euler graph if and only if it can be decomposed into circuits.
(b) Define and illustrate with suitable examples the terms eccentricity and centre of a graph G.
(c) Prove that if A and B are independent events, then A^c and B^c are also independent events. 6+(2+2)+6
4. (a) State and prove the generalized Pigeonhole Principle.
(b) There are six processes to be assigned to three processors. In how many ways can you distribute these processors over processors such that no processors will be idle? A processor can accept multiple processes but a single process is to be allocated by a single processor only.
(c) What is an Adjacency Matrix of a graph G? State two of its characteristics. What is a Path Matrix? (2+4)+4+(2+2+2)